

Computer Science Department
Software Engineering & Business Analysis

Bachelor's Thesis

MovieCrush

Movie Tracking Platform with Personalized
Recommendations, Detailed Ratings, and Social Features

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Submission date

16 June 2026

TODOS

1.4 Boxes	11
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- [Personnal todo before marking this thesis as final](#)

Academic Integrity Statement

I, undersigned, hereby declare that this capstone project is the result of my own work.

- All ideas, data, figures and text from other authors have been clearly cited and listed in the bibliography.
- No part of this project has been submitted previously for academic credit in this or any other institution.
- All code, diagrams, and third-party materials are either my original work or are used with permission and properly referenced.
- I have not engaged in plagiarism or any form of academic dishonesty.
- Any assistance received (e.g. from peers, tutors, or online forums) is acknowledged in the acknowledgements section.

I understand that failure to comply with these declarations constitutes academic misconduct and may lead to disciplinary action.

Place, date Kyiv, 26.05.2026

Signature _____

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Thank you to absolutely everyone who came into my life during these years and made it the best it could be - in such difficult times.

Abstract

The abstract serves as a concise summary of your entire thesis, encapsulating key elements on a single page such as:

- *General background information*
- *Objective(s)*
- *Approach and method*
- *Conclusions*

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1 | Introduction

Your introduction serves to introduce the topic of your Bachelor thesis and to arouse the reader's curiosity with an overview. Why it is important and how it is structured, we explain here.

You can consider an introduction as a teaser for your bachelor thesis. You arouse interest and give a foretaste by presenting your motivation, your method and the state of research in your introduction.

Convince your examiners already in the introduction that your Bachelor thesis will be exciting. If your professor starts reading your thesis with anticipation and interest, the chances of getting good grades are higher.

Pay particular attention to the following in your introduction:

- **Introduce the topic** - What characterizes the topic?
- **Introduce the goal** - What do you want to achieve with your thesis?
- **Make the reader curious** - What motivates the reader to read on?
- **Describe the relevance** - Why is this bachelor thesis scientifically relevant?

The introduction should have the following content:

- **Initial situation presentation of the topic** - You introduce the topic with an exciting 'bait'. You provide initial information on the topic and the object of research and explain the current state of research.
- **Relevance of the topic motivation** - You justify the relevance of your topic (scientifically) and place it in the context of your field. In addition, it is often required that you disclose your personal motivation.
- **Objectives** - Your introduction should clearly state what the goal of your paper is and what outcome you hope to achieve upon completion of the bachelor thesis.
- **Method** - You explain the approach and justify the choice of method.
- **Structure of the Bachelor's thesis** - Finally, you give the reader a general overview of your Bachelor's thesis by explaining the structure, showing the red thread and how the research question is answered.



Welcome to the template's introductory chapter! Instead of boring you with lorem ipsum, here's a quick guide to what you can do in Typst and, more specifically, in this template.

Need more? Check out the [Guide to Typst](#).

1.1 Basic markup

Typst lets you create bold, italic, or monospaced text with ease. You can also sprinkle in equations like $e^{i\pi} + 1 = 0$ or even inline code like `fn main() { println!("Hello, World!") }`. And because life is better in color: pink, blue, yellow, orange, green, and more! **Boldly** colorize!

You can also write numbered or unnumbered lists:

- First item
- Second item
 1. First Subitem

- 2. Second Subitem
- Third item

Need equations? Sure! They look great as blocks too:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

1.2 Images

As they say, a picture is worth a thousand words. Let's add one:



Figure 2: Project logo

1.3 Tables

Tables are great for organizing data. From simple to complex, Typst handles them all:

Name	Age	City
Albert Einstein	25	Bern
Marie Curie	22	Paris
Isaac Newton	30	London

Table 1: Simple table

[31:27]			[24:20]		[19:15]	[14:12]	[11:7]		[6:0]	
funct5		aq	rl	rs2	rs1	funct3	rd	opcode		
5				5	5	3	5	7		

Table 2: Complex table

1.4 Boxes

Highlight key points with these fun boxes (and more):



Infobox: For highlighting information.



Ideabox: Share a brilliant idea.



Warningbox: Proceed with caution!



Firebox: This is 🔥 !



Rocketbox: Shoot for the moon!



Todobox: Just do it!

TODO

Personnal todo before marking this thesis as final

1.5 Citations, Acronyms and Glossary

Add citations with @ like [1] or [1, p.7ff] (stored in **/tail/bibliography.bib**).

Acronym terms like **Infotronics (IT)** expand on first use and abbreviate after **IT**. Glossary items such as **Rust Programming Language (Rust)** can also be used to show their description as such: Rust is a modern systems programming language focused on safety, speed, and concurrency. It prevents common programming errors such as null pointer dereferencing and data races at compile time, making it a preferred choice for performance-critical applications.. Acronyms and glossary entries auto-generate at the document's end (defined in **/tail/glossary.typ**).

1.6 Code

Besides writing inline code as such `fn main() { println!("Hello World") }` you can also write code blocks like this:

```

1 fn main() {
2     let ship = Starship::new("USS Rustacean", (0.0, 0.0, 0.0));
3     let destination = (42.0, 13.0, 7.0);
4     let warp = ship.optimal_warp(ship.distance_to(destination));
5
6     println!("👉 {} traveling to {:?} at Warp {:.2}", ship.name, destination,
7 warp);
8     if warp <= 9.0 {
9         println!("🚀 Warp engaged!");
10    } else {
11        println!("⚠ Warp failed!");
12    }
13 }
```

Listing 1: First part of the USS-Rustacean code

or directly from a file

```

1 struct Starship {
2     name: String,
3     position: (f64, f64, f64),
4 }
5
6 impl Starship {
7     fn new(name: &str, position: (f64, f64, f64)) -> Self {
8         Self {
9             name: name.into(),
10            position,
11        }
12    }
13    fn distance_to(&self, dest: (f64, f64, f64)) -> f64 {
14        ((dest.0 - self.position.0).powi(2)
15         + (dest.1 - self.position.1).powi(2)
16         + (dest.2 - self.position.2).powi(2))
17        .sqrt()
18    }
19    fn optimal_warp(&self, distance: f64) -> f64 {
20        (distance / 10.0).sqrt().min(9.0)
21    }
22 }
```

Listing 2: Second part of the USS-Rustacean code from `/resources/code/uss-rustacean.rs`

1.7 Context Problem

Haute École d'Ingénierie (HEI) Rust Rust programs

[1], [1, p.7ff]

```

fn main() {
    println!("Hello World!");
}
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

1.8 Objectives

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

1.9 Structure of this report

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

2 | Research

2.1 Research Methodology

To validate the problem and understand user needs before building MovieCrush, three complementary research methods were used:

- **Quantitative survey** - a structured questionnaire distributed to 262 respondents to measure user behavior, pain points, and feature demand at scale.
- **Qualitative user interviews** - five in-depth interviews conducted to uncover motivations, frustrations, and unmet needs that surveys cannot capture.
- **Competitor analysis** - a systematic review of existing film tracking platforms to identify gaps in the market and opportunities for differentiation.

All three methods were applied before development began, ensuring that product decisions were driven by real user needs and market evidence rather than assumptions.

2.2 Quantitative Research: User Survey

The survey consisted of 5 blocks of questions:

- **Demographics:** age, content language, profession
- **Digital habits:** streaming subscriptions, monthly spend
- **Viewing habits & pain points:** how many films per week, how they currently track films, what they like/dislike about existing apps
- **Feature validation:** interest in MovieCrush features (AI recommendations, Wrapped, Soulmate, detailed ratings, challenges)
- **Monetization:** attitude to ads, willingness to pay for premium (3 USD/month)

The full survey is available [here](#). Raw responses from all 262 respondents are available in [Google Sheets](#).

2.2.1 Survey Distribution & Incentive

To maximize participation, the survey was distributed via Slack to the KSE student community. To motivate respondents, 4 cinema tickets were raffled among participants - 2 winners, 2 tickets each for any film and any date of their choice.

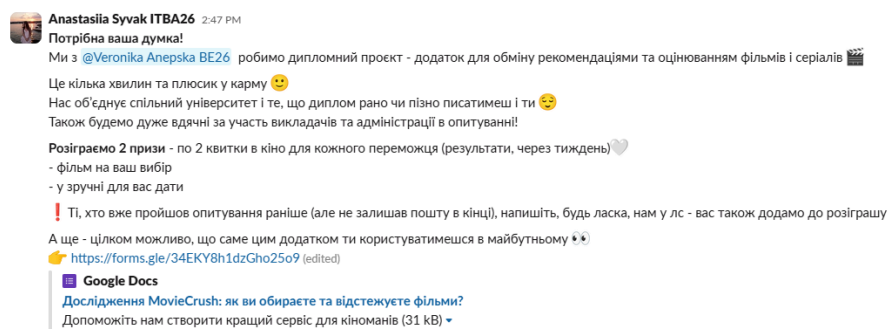


Figure 3: Survey announcement in KSE Slack

The announcement generated strong engagement - 43 reactions, which is significantly higher than typical KSE Slack posts that receive 2-6.

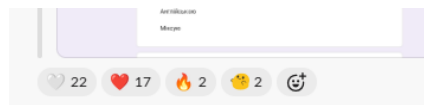


Figure 4: Community reactions - 43 emoji responses to the announcement

Both winners were contacted and confirmed receipt of their cinema tickets.

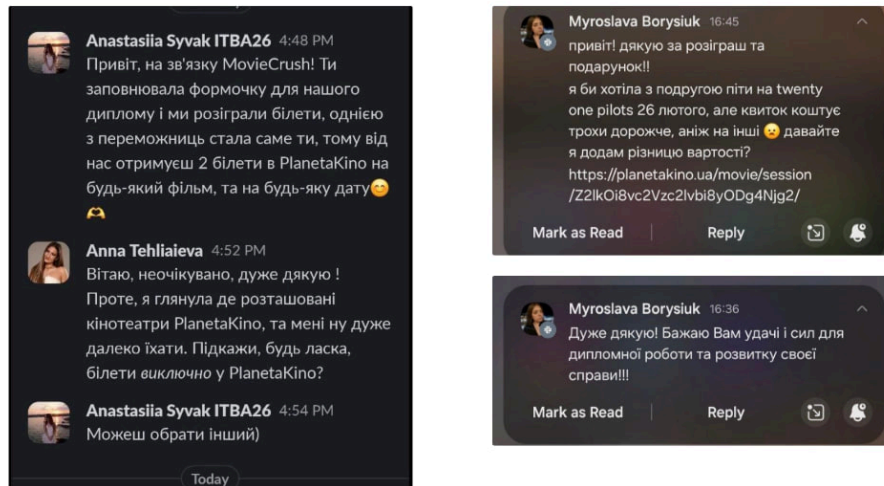


Figure 5: Winner confirmation messages

2.2.2 Demographics

The survey collected 262 responses. The majority of respondents - **90.5%** - were aged 18-24, which reflects the author's social circle and the primary target audience for MovieCrush. **82.7%** have active streaming subscriptions, confirming high purchasing power and an established habit of paying for digital content.

Вкажіть ваш вік

262 responses

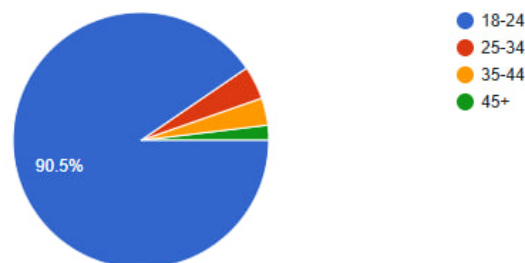


Figure 6: Age distribution of survey respondents

In terms of content language, **42.4%** consume content in a mix of Ukrainian and English, **36.3%** primarily in Ukrainian, and **21.4%** in English only - confirming that the app must

support both languages. Some respondents mentioned russian-language content. The author's position is clear: russian will not be supported. MovieCrush targets Ukraine, Europe, and the US - not the CIS market.

Якою мовою ви переважно споживаєте кіноконтент?

262 responses

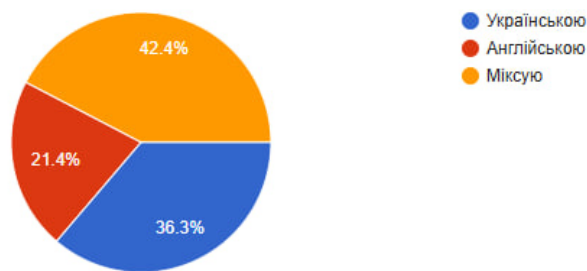


Figure 7: Language in which respondents consume film content

2.2.3 Viewing Habits & Pain Points

43.1% of respondents watch 2-5 films or series per week - the core active audience that needs tools for tracking and recommendations.

Скільки фільмів (або серій) ви переглядаєте в середньому за тиждень?

262 responses

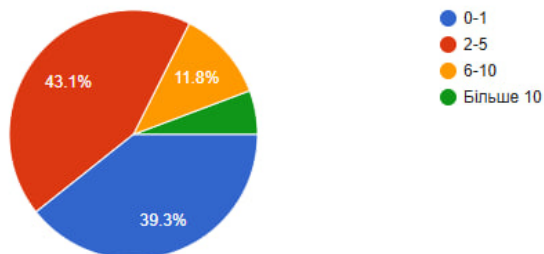


Figure 8: Average number of films or series watched per week

When asked how they currently track films, **49.8%** said they use phone notes - a clear signal that no proper tool exists. Only 19.8% use dedicated services like Letterboxd or IMDb, suggesting low awareness or poor fit with existing solutions. Additionally, 7.6% use TikTok or Instagram to save films - a new behavioral pattern no existing app addresses. 24.9% do not track films at all, representing untapped latent demand.

Як саме ви зараз запам'ятовуєте/відстежуєте фільми, які хочете подивитися або вже подивилися?		
Не відстежую	59	24.9%
Нотатки в телефоні	118	49.8%
Використання наших конкурентів	47	19.8%
Збережені в тік токі/інстаграм	18	7.6%
Списки в стрімінгах (Netflix)	39	16.5%

Figure 9: How respondents currently track films they want to watch or have watched

2.2.4 Competitor Pain Points

When asked what bothers them about existing services like IMDb and Letterboxd, respondents highlighted recurring problems:

- **Poor recommendation algorithms** - platforms suggest popular content, not personalized picks
- **No TV series or anime** - Letterboxd ignores a huge segment of content
- **No statistics** - users want a Spotify Wrapped equivalent for films
- **Outdated interface** - IMDb is losing younger users due to poor design
- **No Ukrainian language** - a barrier for the mass Ukrainian audience
- **Fragmented lists** - content saved across multiple platforms with no unified view

At the same time, users value what currently works: trust in IMDb ratings, seeing what friends watch, viewing diaries, and watchlists. These became the foundation for MovieCrush's core feature set.

2.2.5 Where Users Look for Recommendations

83.6% of respondents look for recommendations on social media (Instagram, TikTok, Twitter), and **76%** rely on friends. Only **7.3%** use specialized apps like Letterboxd or TV Time. This confirms that existing platforms are failing at discovery.



Figure 10: Where respondents usually look for recommendations on what to watch

58% actively discuss films with friends and family, and **37%** do so occasionally - confirming strong demand for social features.

2.2.6 Feature Validation

Respondents were asked which features they would be willing to pay for. The top two were the **hybrid recommendation system** (42%) and **personal statistics/Wrapped** (35.9%). Social features like Soulmate (14.5%) showed lower willingness to pay but high viral potential - making them a strong fit for the free tier.

Якщо основний функціонал (трекінг, списки, стрічка) буде безкоштовним, за які ДВІ додаткові функції з цього списку ви б були готові платити окремо?

[Copy chart](#)

262 responses

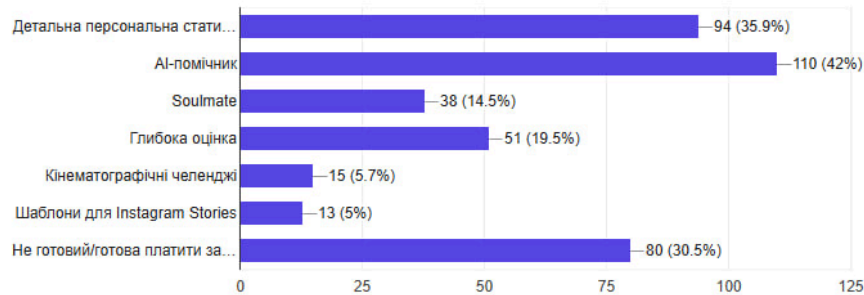


Figure 11: Features respondents would be willing to pay for separately

It is worth noting that during the survey, the planned feature was described as an “AI assistant.” In the final implementation, this became a hybrid recommendation system combining ALS collaborative filtering with TMDb Discover for candidate generation, and Gemini for re-ranking - a more technically sound solution to the same user need.

The personal statistics feature (Wrapped) received an average rating of **4.24 out of 5** - the highest of all features tested. Full rating distribution is available in the Appendix.

2.2.7 Social Features

When asked which social feature mattered most, **48.9%** chose seeing what their friends are watching and rating, **38.5%** wanted shared lists, and **35.1%** wanted to compare tastes with friends. This strongly validates the social layer of MovieCrush.

Яка з цих соціальних функцій була б для вас найважливішою? (Оберіть 1-2)

[Copy chart](#)

262 responses

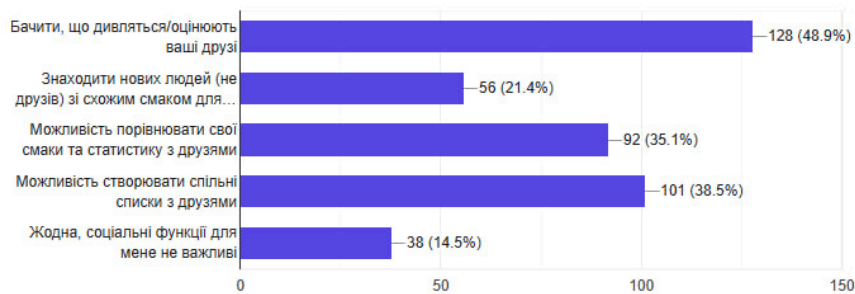


Figure 12: Most important social features according to respondents

2.2.8 Monetization Insights

67.6% of respondents would accept ads in a free version, validating the freemium model. **23.3%** said ads would bother them even in a free version - this segment is the guaranteed base for premium conversion.

Як би ви ставились до реклами в такому додатку?

262 responses

[Copy chart](#)



Figure 13: Respondent attitude toward ads in the app

On pricing, 75.5% of respondents would consider paying 3 USD or more per month for a premium package including AI recommendations, extended statistics, Soulmate search, and no ads. Only 9.9% said they would not pay at all.

Якщо б преміум-підписка включала: відсутність реклами, AI-рекомендації, Soulmate-пошук, розширену статистику - за яку максимальну щомісячну ціну ви б розглянули її придбання?

262 responses

[Copy chart](#)

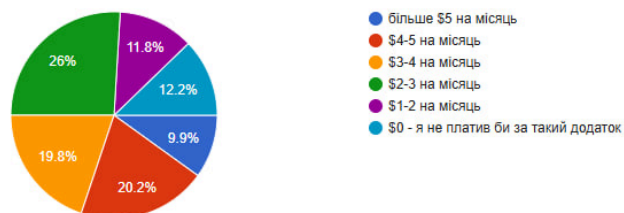


Figure 14: Maximum monthly price respondents would consider for a premium subscription

2.2.9 Open-ended Responses

The final open-ended question asked what would motivate users to switch to a new film app. The most frequent themes across 262 responses were:

Theme	Mentions
AI / Personalized recommendations	25+
Clean and beautiful interface	20+
Statistics / Wrapped	15+
Social features (friends, lists)	15+
Large library / anime / series	12+
Support for Ukrainian product	10+
Unique features (Soulmate, challenges)	8+
Price / free access	7+
Niche content	5+

Table 3: Most frequent themes in open-ended responses

2.2.10 Ideal Customer Profile

Based on all survey findings, the target segment is:

- **Age:** 18-24 years old (90.5% of respondents)
- **Status:** Students (32.3%) and young IT professionals (17.7%)
- **Language:** Mix of Ukrainian and English (42.4%)
- **Viewing habits:** 2-5 films/series per week (43.1%)
- **Spending:** up to 25 USD/month on subscriptions
- **Pain point:** Using phone notes to track films (49.8%)
- **Wants:** hybrid AI recommendations (42%) and personal statistics (35.9%)



Strong Product-Segment Fit for: “Ukrainian tech-oriented youth aged 18-24, actively consuming content, unsatisfied with existing solutions, and willing to pay 3 USD/month for a quality personalized experience.”

2.3 Qualitative Research: User Interviews

Five in-depth interviews were conducted with target users before development began. Recordings are available at: [Telegram channel](#).

2.3.1 Methodology

In-depth interviews were chosen over surveys because open-ended questions force respondents to formulate their own opinions rather than selecting from predefined options. Individual format allows the interviewer to follow unexpected threads and uncover insights that are not on the surface. Group interviews were avoided to prevent social influence on answers.

Respondents were selected to represent different life situations: two students, one working professional, and two family respondents. The only selection criterion was an interest in watching films or series. Unlike the quantitative survey - where 90.5% of respondents were aged 18-24 - the interview participants represented a broader age range. This was intentional: to capture different behavioral patterns and avoid confirming only the assumptions of one demographic group.

Each interview was conducted at a convenient time and location for the respondent. All participants gave verbal consent to recording and further use of the data.

2.3.2 Interview Findings

Time spent searching varies dramatically. Some respondents find something in 5 minutes, others spend up to 3 hours. Regardless of the actual time, all respondents considered the search process cognitively exhausting:



“If within 2-3 minutes I can’t find anything - the desire just passes and I don’t watch anything at all.”

Phone notes are universally used but universally disliked. Every respondent who keeps lists uses phone notes or saved TikToks. All of them described this as inconvenient:



“It’s not convenient because there are too many actions - you always need to adjust it, add things, delete things.”

TikTok and Instagram edits are the primary discovery tool. Respondents do not rely on streaming algorithms to discover new content. Instead, they use short video edits to decide whether a film matches their mood before committing to watch it.

Friends’ recommendations are trusted more than ratings. IMDb ratings were specifically criticized for being misleading:



“I don’t like IMDb ratings - they often spoil my impression of a film. I would rate it 10, but IMDb shows 3.”

Filters are the most requested feature. When asked what they want in an app, almost every respondent mentioned filters:



“The more filters, the better - the faster you can find what you want.”

A successful recommendation creates a strong emotional reaction. When asked how they feel after finding the perfect film, respondents used words like “euphoria” and “delight” - confirming that the emotional payoff of a good recommendation is significant.

2.3.3 User Persona

Based on the interviews, the following persona was developed:

Name	Anna
Age	25
Status	Junior specialist, active schedule
Platforms	Netflix, HD Rezka, YouTube
Discovery	TikTok edits, Instagram, friends
Tracking	Phone notes - finds it inconvenient
Pain point	Spends too much time searching, recommendations don't match expectations
Motivation	Emotional relief after a busy week
Key request	Better filters, see what friends watch

Table 4: User persona derived from qualitative interviews

2.3.4 Implications for MovieCrush

The interviews directly shaped the following product decisions:

- **Mood-based filters** - respondents choose films by mood, not genre alone
- **Social feed** - seeing what friends watch is more trusted than any algorithm
- **Detailed ratings with comments** - numeric ratings are not trusted without context
- **Simple onboarding** - “the more actions required, the less likely people are to use it”
- **Watchlist and tracking** - replacing phone notes with a proper tool was a universal need

2.4 Competitor Analysis

2.4.1 Overview of Existing Solutions

2.4.2 Competitor Gap Analysis

The table below summarizes the key weaknesses identified in competing platforms and how MovieCrush addresses each of them.

Competitor	Key Weaknesses	How MovieCrush Addresses Them
Letterboxd	1. Films only - no TV series support 2. No personalized recommendations 3. Limited social interaction (likes and comments only)	1. Unified catalog: films, series, and individual episodes 2. Hybrid recommendation system: ALS collaborative filtering with Gemini re-ranking 3. Deep social layer: Soulmate, detailed ratings with comments, Best Actor voting
TV Time	1. Poor UX and outdated interface 2. Weak language support 3. Cast-only - no directors or screenwriters 4. Poor recommendations 5. No direct user monetization (revenue from data only) 6. Friends' ratings not visible	1. Modern, intuitive UI/UX built as a core priority 2. Full Ukrainian language support 3. Complete credits: actors, directors, screenwriters 4. Personalized recommendations via ALS + Gemini 5. Direct user monetization through premium subscription 6. Social feed showing friends' ratings and activity
IMDb	1. No social features (no friends, no communities) 2. Overloaded, cluttered interface 3. Tied to Amazon ecosystem 4. No personal analytics beyond basic counters	1. Social layer as a core product feature: follows, Soulmate 2. Clean, user-experience-focused design 3. Independent service with email-based registration 4. Deep personal analytics via Wrapped
Moviebase	1. No social features 2. No gamification 3. Analytics locked behind pay-wall 4. Narrow geographic reach (53% Brazilian audience)	1. Soulmate and social ratings as core free features 2. Gamification planned for future development 3. Wrapped analytics available to all users 4. Designed for Ukrainian, European, and US markets

Table 5: Competitor gap analysis and MovieCrush solutions



Core thesis: MovieCrush combines the strengths of existing platforms while eliminating their key weaknesses, and introduces unique social features.

2.5 Research Limitations



These results reflect a non-representative sample - 89.9% of respondents were aged 18-24. We found **Product-Segment Fit**, not full Product Market Fit.

The following limitations should be considered when interpreting these results:

1. **Non-representative sample** - 89.9% of respondents were aged 18-24, which correlates with the author's age and social circle.
2. **Narrow demographic** - results reflect the views of students in technical and economic fields only.
3. **Conclusion:** We cannot claim full Product Market Fit for the entire market. Instead, we found **Product-Segment Fit** for the segment of "youth aged 18-24, students, technical specializations." Validation on adjacent segments (25-34, non-IT professions) remains a goal for future work.

2.6 Domain-Specific Analysis

2.6.1 Value Chain

Understanding who contributes value in the film tracking domain helps clarify MovieCrush's dependencies and strategic position.

Participant	Role	Impact on MovieCrush
Studios / Rights holders	Own rights to films	Posters and metadata used via TMDb API - direct usage without official API carries DMCA risk
Data aggregators (TMDb)	Collect and maintain meta-data: cast, descriptions, posters	Critical dependency. TMDb is the primary data source - open, well-maintained, and best suited for localization
Streaming services (Netflix, MEGOGO)	Final content providers	Their libraries define what users can watch. Their own recommendation engines are direct competitors
Competitors (Letterboxd, TV Time)	Social film tracking platforms	Set UX standards, monetization models, and user expectations

Table 6: Value chain analysis for the film tracking domain

2.6.2 Technical Challenges

Building a film tracking platform involves several domain-specific technical challenges that shaped architectural decisions in MovieCrush:

- **AI recommendation complexity:** Unlike music, where taste is often genre-based, cinematic taste is **multimodal** - a person may love a specific director, cinematographer, actor, atmosphere, or even sound design rather than a genre. This makes building a recommendation model significantly harder. MovieCrush addresses this with a hybrid approach: ALS collaborative filtering learns from behavioral patterns across users and has priority as the main source of candidates. TMDb Discover serves as an additional candidate source. Google Gemini then re-ranks the combined pool based on the individual user's watch history and ratings - capturing taste signals that a simple genre filter cannot.
- **Cold start problem:** New users have no watch history, making collaborative filtering impossible. MovieCrush solves this with a dedicated onboarding flow: users select favorite actors, then receive a curated set of films which they mark as watched or not watched and rate. The system analyzes content type and genre signals from these ratings to seed initial recommendations via TMDb Discover.
- **TMDb caching for analytics:** MovieCrush maintains a local PostgreSQL cache of TMDb metadata (title, poster, genres, cast, director) that is populated when a user marks a film as watched or adds a rating. This cache is used for Wrapped statistics and recommendation building to avoid excessive TMDb API calls during computation-heavy operations. For regular browsing, the app fetches data directly from TMDb with

a short-lived in-memory LRU cache to reduce redundant requests within the same session.

2.6.3 Legal & Economic Constraints

- **Copyright on visuals:** Using posters and metadata for informational purposes is generally accepted when sourced through official APIs like TMDb. Direct commercial use without proper licensing carries legal risk. MovieCrush uses only TMDb-provided assets within their terms of service.

3 | Design

In this section you turn your requirements into a concrete engineering blueprint. You'll justify every major architectural choice, visualize structure with C4 diagrams for the first three layers, and map out your runtime topology so that peers can understand—and you can defend—every aspect of your system.

1. Clarify how functional and non-functional requirements drive your high-level architecture
2. List each architectural decision (for example, “We chose microservices to enable independent scaling and deployment”) and explain why it best meets your goals
3. Include a C4 Context diagram showing your system in its environment (users, external systems, data sources)
4. Include a C4 Container diagram breaking the system into deployable units (APIs, web front end, background workers, databases) and annotate communication styles and protocols
5. Include a C4 Component diagram for your core container(s), illustrating key modules, services or libraries and their interactions
6. Describe your deployment topology: physical or cloud hosts, network zones, load-balancing, failover and backup strategies
7. Summarize your technology stack, mapping each tool or framework back to a specific container or component and noting any trade-offs (performance, community support, learning curve)
8. Outline how data flows through the system—including storage models, messaging patterns or API contracts—and note any schema or interface versioning plans
9. Address cross-cutting concerns (Security, Logging, Monitoring, Scalability) and show where they sit in your topology

By walking through Requirements → Decisions → Diagrams → Topology, your Design section becomes a rigorous, evidence-backed foundation for the implementation that follows.

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3.1 Section 1

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3.2 Section 2

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3.3 Conclusion

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4 | Implementation

In this section you translate your component-level designs into working code and systems. Focus on the C4 Component layer and on the details needed to show how your design was realized. Include only the most important code snippets that illustrate key patterns or algorithms, rather than full listings.

1. *Describe the development methodology (for example, Agile or test-driven development) used to guide your implementation*
2. *Explain any prototyping or iterative strategies you applied to refine components before full-scale coding*
3. *Summarize coding standards, naming conventions and architectural patterns followed in your codebase*
4. *Present critical code snippets or configuration templates that highlight how core components were implemented (for example, key classes, interfaces or algorithms)*
5. *Detail your testing approach and quality assurance measures (unit tests, integration tests, coverage metrics)*
6. *Note any performance optimizations or profiling results for components that were bottlenecks*
7. *Outline your deployment and configuration management process for component artifacts (containerization, CI/CD pipelines)*
8. *Highlight documentation deliverables (API references, inline comments, architecture decision records) that support future maintenance*

This section demonstrates how each component specification becomes actual, maintainable code—closing the loop from design to implementation.

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4.1 Section 1

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4.2 Section 2

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4.3 Conclusion

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5 | Validation

Validation (Requirements Verification and Testing)

In this section you demonstrate how your implementation satisfies the initial requirements through clear testing methods and concise examples—suitable for a bachelor-level project:

1. *Restate each key functional and non-functional requirement from your Analysis and Design sections*
2. *Describe the testing approach for each requirement (for example, unit tests, manual acceptance checks or scenario walkthroughs)*
3. *Provide concrete test cases or usage examples that show how you verify each requirement in practice*
4. *Summarize actual versus expected outcomes, indicating pass/fail status for each test*
5. *Include brief snippets of test code or sample console outputs to illustrate your procedures*
6. *Note any gaps or deviations and suggest simple fixes or areas for future improvement*
7. *If a feature wasn't intended for specific scenarios (e.g. high-load), omit unrealistic stress tests and clearly document its current limitations*

This focused structure ties every requirement directly to validation results, using examples and methods you can realistically carry out at the bachelor level.

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5.1 Section 1

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5.2 Section 2

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5.3 Conclusion

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6 | Conclusion

Conclusion

In this final section you bring together your work and reflect on its impact. Keep it concise, restating key points without introducing new information:

- 1. Project Summary: Briefly recap objectives, methodology and principal results*
- 2. Alignment with Objectives: Discuss how outcomes meet initial goals, referencing requirements and design aims*
- 3. Lessons Learned and Challenges: Note any obstacles and how they informed improvements*
- 4. Limitations: Acknowledge features or scenarios beyond this scope and clearly state current system boundaries*
- 5. Future Work: Suggest practical enhancements or research directions building on your findings*

Avoid introducing new concepts here; refer readers to the Discussion for deeper analysis.

6.1 Project summary

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6.2 Comparison with the initial objectives

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6.3 Encountered difficulties

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6.4 Future perspectives

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Glossary

Programming Language

Rust – Rust Programming Language: Rust is a modern systems programming language focused on safety, speed, and concurrency. It prevents common programming errors such as null pointer dereferencing and data races at compile time, making it a preferred choice for performance-critical applications. [6](#), [7](#)

University

HEI – Haute École d'Ingénierie [7](#)

IT – Infotronics [6](#)

Bibliography

- [1] S. Zahno *et al.*, “Dynamic Project Planning with Digital Twin,” *Frontiers in Manufacturing Technology*, vol. 3, May 2023, doi: [10.3389/fmtec.2023.1009633](https://doi.org/10.3389/fmtec.2023.1009633).

A | Appendix

AA Survey Materials

AAA Raffle Process

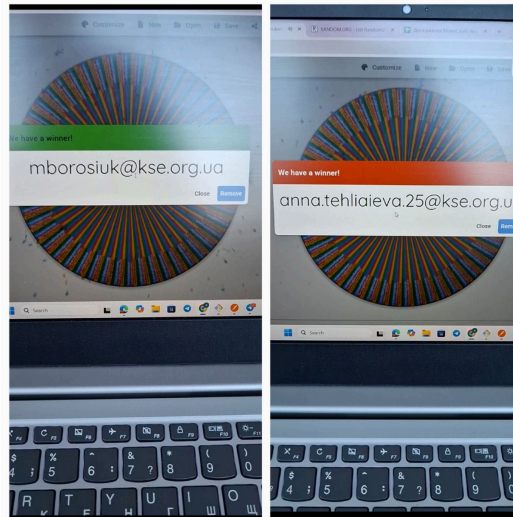


Figure 15: Random wheel spinner used to select raffle winners

AAB Participant Messages

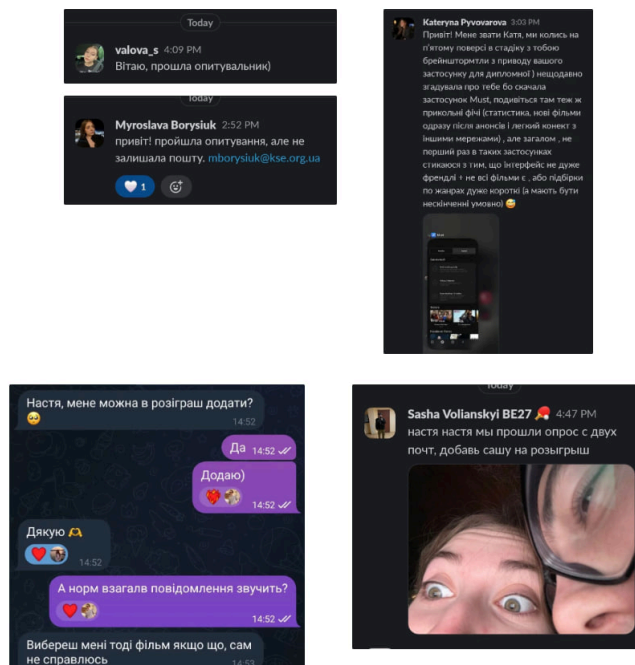


Figure 16: Messages from participants requesting to join the raffle

AB Additional Survey Charts

Note: the tables below are based on 237 responses (an earlier snapshot of the survey). The final response count was 262.

ABA Wrapped Statistics Feature Rating

The Wrapped feature received the highest average rating of all tested features - **4.24 out of 5** - making it the most anticipated feature among respondents. 53.8% gave it the maximum score of 5.

Оцініть, наскільки цікавою для вас була б ця функція:

Детальна персональна статистика (загальна кількість переглянутого контенту, улюблені жанри/режисери, щорічний звіт на кшталт "Spotify Wrapped")

(1 - взагалі не цікаво, 5 - дуже цікаво!)

262 responses

[Copy chart](#)

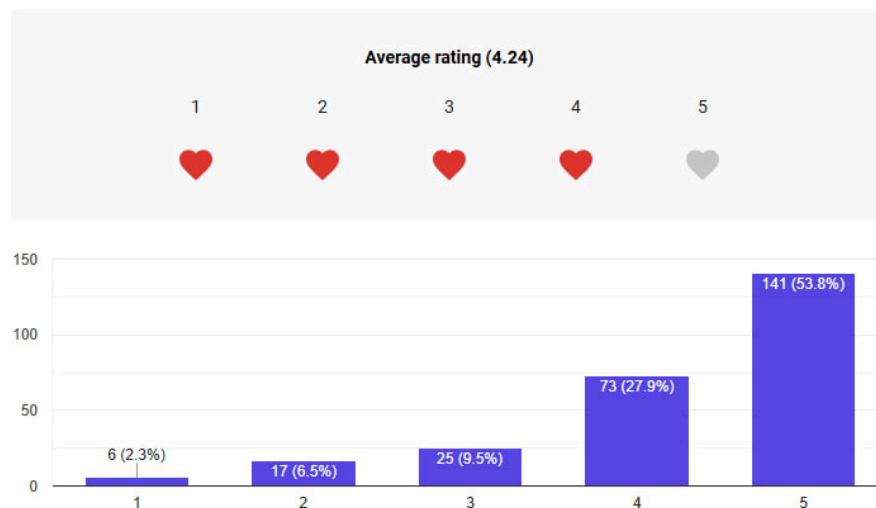


Figure 17: Rating distribution for the Wrapped statistics feature — average 4.24/5

ABB Monthly Spending on Subscriptions

Скільки в середньому ви платите щомісяця за всі свої підписки на контент (стрімінги, музика)?

262 responses

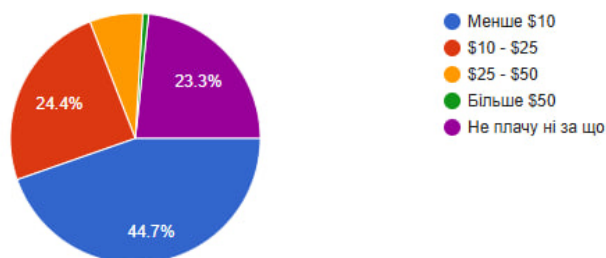


Figure 18: Monthly spending on content subscriptions among respondents

ABC Active Streaming Subscriptions



Figure 19: Active streaming subscriptions among respondents

Підписки на стрімінгових платформах	Кількість людей	Відсоток людей
Немає жодних підписок	41	17.3%
Є підписки	196	82.7%

Figure 20: Streaming subscription summary - 82.7% have at least one active subscription

ABD Respondent Profession & Field of Activity

Segmenting professions was challenging because 89.9% of respondents were aged 18-24, meaning most were students. Some listed only “student”, others listed their specialization, and those already working listed only their job title. To handle this, three separate breakdowns were created:

1. **General table** - all segments combined
2. **Profession table** - “Student” as a category plus job titles of those already working
3. **Field of activity table** - students filtered by their area of study, regardless of employment status

Of the respondents: 69 are not working, 51 did not answer, and approximately 117 indicated a profession (though some may have listed their future profession rather than a current job).

Ваша професія/сфера діяльності	
Пусті значення	47
Специфічна відповідь	4
Студент	46
ІТ	34
Аналітика	17
Економіка	18
Інші професії	14
Менеджмент	7
Аудит	8
Студент, Психологія	3
Право	4
Психологія	7
Викладач	8
Студент, ІТ	11
Студент, Економіка	6
Студент, Право	3

Ваша професія	
Пусті значення	47
Специфічна відповідь	4
Студент	69
ІТ	34
Аналітика	17
Економіка	18
Інші професії	14
Менеджмент	7
Аудит	8
Право	4
Психологія	7
Викладач	8

Сфера діяльності	
Пусті значення	47
Специфічна відповідь	4
Студент	46
ІТ	45
Аналітика	17
Економіка	24
Інші професії	14
Менеджмент	7
Аудит	8
Право	7
Психологія	10
Викладач	8

Figure 21: Three-way breakdown of respondent profession and field of activity